

The Proto-Spinor: Conditional Spinorial Closure and the q79 Interface

Corrected sixth edition

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Abstract

We isolate the part of the proto-spinor construction that is mathematically forced after its geometric premises are stated. A world-in-world field is a section of $\text{Hom}(TP, TI)$ for oriented rank-three bundles; its nine local components do not multiply manifold dimensions. If closure data must retain the nontrivial loop class in $SO(3)$, the faithful continuous double-cover carrier is spinorial and is represented through $\text{Spin}(3) \cong SU(2)$. This is a conditional lifting theorem, not a derivation of internal dimension three or of physical fermions from admissibility alone. For the selected q79 degree-three spectral cover we record the exact trace split of ranks $1 + 2$, the common $1 + 2 + 3$ carrier, and the local binary-dihedral lift of signed sheet monodromy. The shared circle is now a specified pullback of one universal flat $\mathbb{Z}_{64}$ differential line, not merely an isomorphic fiber. Its determinant, CLN, root-plane, and finite-Hessian actions intertwine exactly. The BHT and Hori fiber transforms share one orthogonal complex structure J_{FM} , and the binary sheet group has exactly two conjugate $\pm i$ phase-root lifts for which $(\pm i J_{\text{FM}})^2 = I$. Strict global Spin closure remains equivalent to a branch-relator obstruction calculation; the closed result is SpinC and finite symbolic. Circle, lens, and nil remain typed phase, finite-transport, and anchoring roles. Lorentzian spinors, masses, and particle assignments remain downstream realization problems.

Revision note for this edition

Supersedes. *The Proto-Spinor: Conditional Spinorial Closure and the q79 Interface*, version 5.

Reason. Version 5 stopped at a local Dic_3 lift and an untyped shared line. It did not include the later universal differential-line pullback, the common BHT–Hori Clifford operator, or the exact binary SpinC return.

Resolution. Version 6 retains the conditional double-cover theorem and adds the flat q79 shared-line theorem, finite Hessian naturality, the exact $J_{\text{FM}}^2 = -I$ polarization, and the two conjugate $\pm i$ phase-root lifts.

Retained result. Spinorial lifting is necessary when an oriented rank-three carrier must retain the nontrivial $SO(3)$ loop class.

Remaining boundary. Strict global q79 Spin, the physical nonflat FM/HYM lift, the strain-to-q79 continuum intertwiner, and a Lorentzian particle/action completion remain open gates.

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1 Status and dependencies

This paper is a conditional carrier theorem. Its conclusions depend on the following data, none of which follows from a bare projection map:

1. oriented Euclidean rank-three bundles TP and TI over a declared base B ;
2. a comparison field $Q_{\text{WW}} \in \Gamma(\text{Hom}(TP, TI))$;
3. an $SO(3)$ frame or holonomy bundle whose nontrivial loop class is to be retained by the state description;
4. for the q79 specialization, the selected degree-three spectral cover, its trace map, and its sheet monodromy; and
5. for a physical fermion interpretation, a separate Lorentzian spin or Spin^c completion, action, observable map, and normalization.

The terms “forced” and “necessary” below always mean necessary relative to these premises. They do not mean that MTT has selected the premises uniquely.

2 World-in-world comparison data

Definition 2.1 (Comparison field). Let $TP, TI \rightarrow B$ be oriented Euclidean vector bundles of rank three. A world-in-world comparison field is

$$Q_{\text{WW}} \in \Gamma(\text{Hom}(TP, TI)).$$

At $b \in B$, choices of oriented orthonormal frames represent $Q_{\text{WW}}(b)$ by a matrix in $\text{Mat}(3, \mathbb{R})$.

Proposition 2.2 (Dimension and component bookkeeping). *The total space of $TI \rightarrow B$ has dimension $\dim B + 3$. If $\dim B = 3$, its dimension is six. The number nine is instead the fiber dimension of $\text{Hom}(TP, TI)$.*

Proof. A rank- r vector bundle over a d -dimensional base is locally $U \times \mathbb{R}^r$ and therefore has total dimension $d + r$. Furthermore, $\text{Hom}(\mathbb{R}^3, \mathbb{R}^3)$ has dimension nine. $\square$

At an invertible, orientation-preserving background, polar decomposition gives $Q_{\text{WW}} = RU$ with $R \in SO(3)$ and U symmetric positive definite. Its linearized carrier has the orthogonal Frobenius decomposition

$$\text{Mat}(3, \mathbb{R}) = \mathfrak{so}(3) \oplus \text{Sym}(3, \mathbb{R}), \quad 9 = 3 + 6.$$

After a flag is selected, the symmetric part further decomposes as

$$\text{Sym}(3, \mathbb{R}) = \mathbb{R}I_3 \oplus \mathcal{D}_0 \oplus \mathcal{O}, \quad 6 = 1 + 2 + 3,$$

where $\mathcal{D}_0$ is traceless diagonal and $\mathcal{O}$ is symmetric off-diagonal. The latter split depends on the selected flag; only the $1 + 5$ trace/traceless split is invariant under the full orthogonal group.

Consequently

$$1 + 3 \times 3 = (1 + 3) + (1 + 2 + 3) = 4 + 6 = 10$$

is an exact component identity once an ordering scalar is added. It is not a proof of a ten-dimensional manifold, a $3 + 1$ spacetime, or Lorentzian signature.

3 Conditional spinorial lifting theorem

Let $F^+(TI) \rightarrow B$ be the oriented orthonormal frame bundle. Its structure group is $SO(3)$ and

$$\pi_1(SO(3)) \cong \mathbb{Z}_2, \quad 1 \longrightarrow \{\pm 1\} \longrightarrow \text{Spin}(3) \longrightarrow SO(3) \longrightarrow 1.$$

Definition 3.1 (Neutrality with loop memory). A carrier has neutrality with loop memory when its projected $SO(3)$ frame returns after a closed orientation loop while its state representation still distinguishes the two homotopy classes of such loops.

Theorem 3.2 (Conditional double-cover necessity). *Suppose a continuous state carrier over the $SO(3)$ frame data is required to distinguish the generator of $\pi_1(SO(3))$ and to compose consistently under concatenation of lifted paths. Then an ordinary vector representation of $SO(3)$ is insufficient. The minimal connected covering group on which the nontrivial class remains visible is the universal double cover $\text{Spin}(3) \cong SU(2)$.*

Proof. Every $SO(3)$ representation identifies a loop with its endpoint group element, so both homotopy classes of loops ending at the identity act as the identity. Path lifting to the universal cover assigns the two classes the endpoints $+1$ and -1 . Since $\pi_1(SO(3)) = \mathbb{Z}_2$, the universal connected cover is two-sheeted and no smaller connected cover can retain that class. The universal cover is $\text{Spin}(3) \cong SU(2)$. $\square$

Definition 3.3 (Proto-spinor). Relative to the preceding hypotheses, a proto-spinor is a section of an associated complex rank-two bundle

$$S_I = P_{\text{Spin}} \times_{SU(2)} \mathbb{C}^2,$$

where P_{Spin} is a chosen lift of the relevant oriented frame or holonomy data. The word “proto” records that no Lorentzian Dirac operator, statistics law, or particle ontology has yet been supplied.

The theorem does not prove that an internal world must be three-dimensional. For example, $SO(2)$ already has nontrivial winding, while higher-dimensional orientation groups also possess spin covers. Rank three is a selected realization premise supported here by the comparison carrier and the q79 degree-three data, not a universal dimensional-minimality theorem.

4 Selected q79 trace-split carrier

Let $\pi : C \rightarrow B$ be the selected degree-three spectral cover and set

$$\mathcal{A} = \pi_* \mathcal{O}_C.$$

The unit and trace define

$$p_{\text{cen}} = \frac{1}{3} \mathbf{1} \circ \text{Tr}, \quad \mathcal{A}_0 = \ker \text{Tr}.$$

Theorem 4.1 (Trace-split rank flag). *If $\mathcal{A}$ is finite locally free of rank three and the ground field has characteristic different from three, then*

$$\mathcal{A} \cong \mathcal{O} \oplus \mathcal{A}_0, \quad \text{rank}(\mathcal{O}, \mathcal{A}_0, \mathcal{A}) = (1, 2, 3).$$

Consequently the shared-line carrier

$$\mathcal{H}_{\text{CLN}} = L_{\text{shared}} \otimes (\mathcal{O} \oplus \mathcal{A}_0 \oplus \mathcal{A})$$

has rank six.

Proof. The normalized trace projector is idempotent, has image the unit line, and has kernel $\mathcal{A}_0$. Rank additivity gives $3 = 1 + 2$, and the displayed direct sum has rank $1 + 2 + 3 = 6$. $\square$

This theorem selects a direct trace/trace-zero/full carrier. On a connected cover with transitive monodromy it does not produce a global ordering of three individual sheets.

5 Universal shared differential line

Let $\mathcal{L}_{64}^{\text{univ}} \rightarrow B_{\nabla} \mathbb{Z}_{64}$ be the universal flat line for the primitive character $\chi_1(n) = \exp(2\pi i n/64)$. The unique nontrivial homomorphism

$$h_{S_3} : S_3 \longrightarrow \mathbb{Z}_{64}, \quad h_{S_3}(\sigma) = 32 \epsilon(\sigma),$$

sends even permutations to zero and odd permutations to 32. Pulling $\mathcal{L}_{64}^{\text{univ}}$ back along the q79 sheet classifying map gives $(L_{\text{sh}}, \nabla_{\text{sh}})$. For both admissible odd roots $r \in \{1, 33\}$,

$$\chi_r \circ h_{S_3} = \text{sgn}.$$

Hence L_{sh} and the SpinC determinant sign line are canonically parallel-isomorphic. This equality preserves connection and holonomy; it is stronger than recognizing the same order-two phase pattern.

Theorem 5.1 (Shared-line finite naturality). *The line L_{sh} tensors the trace/trace-zero/full carrier and acts by scalar holonomy. It therefore commutes with all three lane projectors, the root-plane quarter-turn J_{DE} , and the normalized finite Hessian*

$$H_{\text{fin}} = \kappa_{\text{fin}}(I - P_{\text{Haar}}), \quad P_{\text{Haar}} = \frac{1}{6} \sum_{g \in S_3} \rho(g).$$

On the selected two-copy sheet/edge representation,

$$\text{rank } P_{\text{Haar}} = 2, \quad \text{spec}(H_{\text{fin}}/\kappa_{\text{fin}}) = \{0^{\times 2}, 1^{\times 4}\}, \quad H_{\text{TT}} = \kappa_{\text{fin}} I_2.$$

Thus the determinant, CLN, root-plane, and finite-Hessian occurrences are coherent pullbacks of one differential line, with no new fit parameter.

Proof. The determinant statement is the displayed character identity. Scalar holonomy commutes with every sheet operator. Haar averaging is an orthogonal projector, and direct decomposition of the selected representation gives the rank and spectrum. Tensoring by L_{sh} preserves all identities. $\square$

This theorem is flat and finite-symbol exact. The full physical HYM carrier has nonzero characteristic curvature and cannot literally have this flat connection. Its promotion requires a spectral-symbol functor and a parallel continuum Hessian comparison.

6 Signed sheet monodromy and its lift

For a permutation $\sigma \in S_3$, let P_σ be its permutation matrix and define

$$\rho_+(\sigma) = \text{sgn}(\sigma) P_\sigma.$$

Since $\det(\text{sgn}(\sigma) P_\sigma) = 1$, this is a representation $S_3 \rightarrow SO(3)$.

Theorem 6.1 (Local binary-dihedral lift). *The inverse image of $\rho_+(S_3)$ in $\text{Spin}(3)$ is the binary-dihedral group Dic_3 of order twelve. Lifts q_1, q_2 of adjacent transpositions may be chosen so that*

$$q_1^2 = q_2^2 = -1, \quad q_1 q_2 q_1 = q_2 q_1 q_2, \quad (q_1 q_2)^3 = -1.$$

Proof. The signed permutation image is the rotational symmetry group of an oriented triangular frame, isomorphic to the dihedral rotation group of order six. The inverse image of a finite rotation group under the two-to-one map $SU(2) \rightarrow SO(3)$ has twice its order and is its binary counterpart. Standard half-angle quaternion lifts of the two generating rotations satisfy the displayed relations, identifying the inverse image with Dic_3 . $\square$

6.1 Global obstruction

The preceding theorem is not yet a strict global q79 Spin theorem. Let B° be the branch complement and let $\rho : \pi_1(B^\circ) \rightarrow SO(3)$ be the signed sheet monodromy. A strict lift is a homomorphism $\tilde{\rho}$ making

$$\begin{array}{ccc} & \text{Spin}(3) & \\ & \downarrow \tilde{\rho} & \\ \pi_1(B^\circ) & \xrightarrow{\rho} & SO(3) \end{array}$$

commute. It exists exactly when the pulled-back central extension is trivial, equivalently when the relevant w_2 obstruction vanishes. Computationally, one must lift a presentation of

$\pi_1(B^\circ)$ and verify that every relator closes with central sign $+1$. Local braid relations alone do not decide all global relators.

A shared $U(1)$ line can instead participate in a Spin^c cancellation of the mod-two obstruction. That is a distinct theorem and must not be reported as a strict Spin lift.

7 Clifford polarization and exact Spin^c return

On the oriented two-circle exterior algebra with basis $(1, p, q, pq)$, the Poincare-kernel transform is the integral operator

$$J_{\text{FM}} = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

Direct multiplication gives

$$J_{\text{FM}}^T J_{\text{FM}} = I, \quad \det J_{\text{FM}} = 1, \quad J_{\text{FM}}^2 = -I.$$

Thus J_{FM} is an orthogonal complex structure, equivalently one $\text{Cl}_{0,1}$ generator, and

$$P_\pm = \frac{1}{2}(I \mp iJ_{\text{FM}})$$

are complementary complex rank-two polarizations. The same derived operator reproduces the selected BHT and two-circle Hori fiber transforms.

The half-turn character of S_3 cannot itself cancel J_{FM}^2 : an involution maps to 0 or 32 in $\mathbb{Z}_{64}$, while the square roots of 32 are 16 and 48. The correct domain is the binary sheet group Dic_3 already obtained above.

Theorem 7.1 (Binary-sheet Spin^c –Fourier return). *There are exactly two homomorphisms*

$$\tilde{h}_\pm : \text{Dic}_3 \longrightarrow \mathbb{Z}_{64}$$

which send the binary central element to 32 and the two braid generators to (16, 16) or (48, 48). Under either odd character these phases are $+i$ and $-i$, respectively. In the Fourier frame the binary transposition lift is J_{FM} , and therefore

$$U_\pm = \pm iJ_{\text{FM}}, \quad U_\pm^2 = I.$$

The two lifts are complex conjugates with the same determinant sign line. No continuous or discrete fitted parameter is introduced.

Proof. The binary relations force both generator images to be equal square roots of 32; exhaustive solution in $\mathbb{Z}_{64}$ leaves only 16 and 48. The real spinor representation identifies the binary transposition with J_{FM} . Multiplication then gives $(\pm iJ_{\text{FM}})^2 = I$. $\square$

This closes the double return at representation and flat differential-line tier. It does not identify compact phase with physical time, choose one of the conjugate orientations as the observed universe, or supply the nonflat physical FM/HYM correspondence.

8 Circle, lens, and nil as typed carrier roles

In this revision the three labels have the following nonexhaustive roles:

- Circle.** The common line L_{shared} stores phase or holonomy data. It is counted once and is not physical time.
- Lens.** Finite quotient, signed-sheet, or projective transport data. This does not require the global compactification to contain a literal lens-space factor.
- Nil.** Anchoring, upper-triangular transport, or termination data in a selected operator filtration. This does not prove a literal Nil_3 factor.

Thus circle–lens–nil is a carrier and obstruction taxonomy. The auxiliary model $L(3,1) \times \text{Nil}_3$ may realize some roles, but it is not the selected q79 Fu–Yau compactification and the two manifolds are not identified.

Boothby–Wang geometry gives a precise supporting model for this typing. A prequantum circle bundle over $\mathbb{CP}^1$ with first Chern number k has total space $L(k,1)$, while an integral circle bundle over T^2 has a Heisenberg nilmanifold as total space. Thus $L(3,1)$ and Nil_3 are parallel circle-bundle realizations over different bases and curvature classes, not nested spaces [4, 5]. Their curved connections are not the flat q79 root-stack connection. All three may map to the universal classifier $BU(1)_{\nabla}$, but an MTT identification requires selected classifying maps and coherent comparison data.

9 Downstream physical gates

1. A Lorentzian spinor requires a globally hyperbolic base, a spin or Spin^c structure, a tetrad, a compatible connection, and a hyperbolic Dirac operator.
2. A particle interpretation requires a physical state space, statistics, observables, interactions, and an asymptotic or local excitation criterion.
3. A mass is read from a canonically normalized quadratic action or a propagator pole. A positive closure cost alone is not a mass.
4. A Higgs identification requires a selected scalar representation and couplings. A positive Hessian does not imply a unique scalar mode.
5. A discrete spectrum requires a self-adjoint operator with compact resolvent, confining boundary conditions, or another explicit spectral compactness theorem. Nil language alone does not quantize a theory.

Weyl, Dirac, Majorana, and twistor descriptions may be built as conditional encodings once their respective chirality, pairing, real-structure, and conformal hypotheses are supplied. They are not consequences of the rank-three comparison field alone.

10 The same-source interface theorem still required

The local strain carrier and the q79 trace-split carrier both have dimensions $1 + 2 + 3$. Rank equality supplies only a fiberwise linear isomorphism. The needed physical theorem must construct

$$\mathcal{J}_{\text{WW} \rightarrow \text{q79}} : \mathbb{R}I_3 \oplus \mathcal{D}_0 \oplus \mathcal{O} \longrightarrow L_{\text{shared}} \otimes (\mathcal{O} \oplus \mathcal{A}_0 \oplus \mathcal{A})$$

on the selected base and prove:

1. compatibility with transition functions and the selected flag;

2. isometry for the declared strain and HYM inner products;
3. covariant compatibility $\mathfrak{I}\nabla^{\text{WW}} = \nabla^{q79}\mathfrak{I}$; and
4. intertwining of the Hessian, retarded, and overlap operators used by the numerical source packets.

Until these clauses are proved, the local and $q79$ objects are related by a dimension-matched conditional bridge, not an identity. The universal-line and finite SpinC/Fourier theorems above close important internal squares on the $q79$ side; they do not supply this world-in-world-to-continuum map.

11 Relation to the numerical SM program

The current calculation repositories close embedded renormalized-Standard- Model equivalence at the declared one-shared-physical-primitive/profile standard, including the finite matrix, charged Yukawa profile, electroweak, threshold, and precision transport packets. Those are valid at their stated certificate tier. They do not prove that the proto-spinor carrier uniquely selects the measured profile, and they do not close the strict no-knob or unique-observed-branch upgrade. This paper therefore uses the calculations as downstream realization evidence, not as a proof of its upstream premises.

12 Conclusion

The corrected proto-spinor result is narrower and stronger. Under explicit rank-three orientation and loop-memory premises, the double cover is indeed spinorial. The selected $q79$ carrier has an exact $1 + 2 + 3$ trace split and an exact universal flat differential line, an exact Dic_3 lift, and an exact SpinC–Fourier return on its finite carrier. What remains is sharply separated: the strict global Spin obstruction calculation, the physical nonflat FM/HYM lift, the same-source local-to- $q79$ continuum intertwiner, and the Lorentzian action/particle completion.

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